

Plan

ar063 · 11 July 2026 · Draft

This is the **plan** article — the pre-registration and the live experiment queue. Because every demolab page carries the git commit that built it, a plan committed before any experiment has run is provably prior to the results. The policy the overnight shift obeys lives in the night-shift article; the run-by-run record lives in the log.

The goal

We can train the conductance-based E/I network (COBANet) on Spiking Heidelberg Digits, and the first smoke run (exp060) reached 61% on a subset — but it did so **unstably**: the free signed-recurrent dynamics tip the forward pass into NaN divergence on scattered epochs, gradient norms spike into the millions, and the recurrent weights grow without bound. Before any larger claim about what biological structure costs or the gamma rhythm does, the network has to **train reliably**. So the program has exactly one goal, deliberately modest:

Find the minimal recipe that trains a conductance-based E/I spiking network stably on SHD, and attribute each ingredient — integration timestep, weight decay, Dale’s law, the firing-rate regulariser — to the stability it buys.

Bigger results (the cost of biological structure, whether gamma helps classification) wait until this holds. This is the foundation they stand on.

What counts as success. A recipe that trains to completion with:

- *no NaN divergence* — every epoch produces a finite train and test loss;
- *bounded dynamics* — pre-clip gradient norm and recurrent weight norm stay bounded across training, not runaway;
- *decisively above chance* — best test accuracy at or above 25% (chance is 5% on 20 classes), so “stable” does not mean “stably not learning”;

and, alongside the recipe, a table attributing each ingredient to its effect on those stability metrics — what each one is actually for.

The queue

Each entry isolates one candidate ingredient and measures its effect on the stability metrics above (NaN-epoch rate, max pre-clip gradient norm, max recurrent-weight norm), holding the rest of the recipe fixed. Each has a **kill** criterion. Status is read from each run’s outputs, never hand-typed here.

1. exp060 — the failure mode (done). (*status: done*)

- *Result*: the free signed-recurrent ceiling (all four blocks trainable, *-no-dales-law*, dt 1.0, firing-rate regulariser on) trains to 61% on a 1000-sample subset but with intermittent NaN forward-divergence and unbounded W_{ee} growth. This entry defines the problem the rest of the queue solves.

2. exp061 — does finer dt stabilise the integration? (*3 seeds · status: queued*)

- *Hypothesis*: the NaN divergence is exp-Euler stiffness at coarse dt; halving and quartering dt ($1.0 \rightarrow 0.5 \rightarrow 0.25$) drops the NaN-epoch rate toward zero.
- *Kill*: if NaN persists at dt 0.25, coarse integration is not the cause — drop dt from the recipe and look upstream (state clamping).

- *Measures*: NaN-epoch rate and max gradient norm vs dt.
3. **exp062 — is Dale’s law the implicit stabiliser?** (*3 seeds · status: queued*)
- *Hypothesis*: the Dale’s-law constraint (non-negativity clamp) keeps the dynamics in a stable regime; the same recipe **with** Dale’s law trains NaN-free where the free version diverges.
 - *Kill*: if the Dale’s-law net also NaNs at matched settings, the constraint is not what stabilises — the free-vs-constrained difference lies elsewhere.
 - *Measures*: NaN-epoch rate, max weight norm, and best accuracy, free vs constrained.
4. **exp063 — does weight decay bound the free recurrence?** (*status: queued*)
- *Hypothesis*: decay strong enough (sweep $0 \rightarrow 1e-3 \rightarrow 1e-2$) bounds W_{ee} and removes the divergence in the free net. A first pass at $1e-3$ tamed the gradient explosion ($7.5M \rightarrow 900$) but did *not* bound W_{ee} or remove the NaN, so the sweep must reach higher.
 - *Kill*: if even $1e-2$ leaves W_{ee} growing and NaN present, decay is not part of the recipe (it regularises but does not stabilise).
 - *Measures*: max weight norm and NaN-epoch rate vs decay strength.

The firing-rate regulariser is already in the recipe (without it the earlier run blew up to a 526 Hz inhibitory rate); the queue attributes the *remaining* ingredients. If none of dt, Dale’s law, or decay yields a NaN-free recipe, a forward-pass state clamp (bounding voltage and conductance) enters the queue as the fallback — a code change to the shared model, so a last resort.

Amendments

None yet. Any change to a hypothesis or kill criterion after an experiment has run against it is added here as a dated entry, never edited in place.